

Research Article

Marine trophic architecture and hidden ecological connections in the Strait of Magellan: keystone species and ecosystem resilience

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Understanding the ecological implications of species coexistence is central to biodiversity studies and to identify environmental and anthropogenic drivers of ecosystem dynamics, where ecological network analysis offers valuable insights. This study examines the complexity, structure, and potential responses to disturbances of the Strait of Magellan's topological food web. Based on a dataset of 438 predator–prey interactions among 139 trophic species (nodes), the food web was characterized using a two-scale approach. At the network level, descriptors including connectance, degree distribution, and small-world pattern, were used to evaluate overall architecture. At the node level, species' roles were assessed using degree, betweenness, closeness, condensed in a novel Keystone Species Index (KSI), and topological role. Potential resilience to ongoing perturbations was inferred by relating these network- and node-level properties. Overall, the network exhibits low connectance and an asymmetrical degree distribution, with a few species concentrating most interactions. Its small-world pattern, characterized by high clustering and short path lengths, suggests that local disturbances could propagate rapidly. More than half of the species are omnivorous, potentially buffering the system against fluctuations in prey availability. According to the KSI antotopological role results, polychaetes, Fuegian sprat *Sprattus fuegensis* 'sardina fueguina', and Patagonian blenny *Eleginops maclovinus* 'róbalo patagónico', act as central conduits for matter and energy flow, linking benthic and pelagic production to higher trophic levels. The combination of low connectance, an asymmetric degree distribution, and small-world properties suggests that the food web is fragile to perturbations affecting highly connected species, with the potential to trigger trophic cascades. These findings highlight the importance of understanding trophic interactions for effective conservation and ecosystem management and provide new insights for the Strait of Magellan's ecosystem.

Keywords: biodiversity, food web, pelagic and benthic communities, species' role, Sub-Antarctic



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Introduction

The Strait of Magellan, one of the most significant channels in the Magellanic Province (Spalding et al. 2007), serves as a crucial connection between the Pacific and Atlantic Oceans. Its unique high-energy dynamics, influenced by water masses from the Pacific, Atlantic and Southern Oceans (Panella et al. 1991), along with its geological features shaped during the last glaciation (Antezana 1999), make it a region of substantial economic, geopolitical and ecological importance (Ramírez 2005, Ríos et al. 2013, Onetto 2020).

For decades, the Strait of Magellan has supported southern Chile's small-scale fisheries, providing essential resources for local communities (Torres and Ruz 2011). However, these fisheries face increasing risk of overexploitation, as most lack formal management or regulatory safeguards, and concerns are growing over the expansion of unregulated extraction (Sanchez et al. 2011, Almonacid et al. 2018).

Beyond fisheries, multiple human activities intensify pressures on the ecosystem. Oil and gas exploration and extraction have shaped regional development since the early 20th century (Thomas 1949), while the Strait's global role as a maritime corridor sustains heavy shipping traffic. These activities introduce risks such as pollution, habitat degradation, and disruptions to ecological dynamics, with potential cascading effects on marine ecosystems (Crain et al. 2008, Halpern et al. 2008, Butnariu 2022). Preliminary studies suggest that nitrogen enrichment from localized anthropogenic pressures may drive shifts in benthic food-web structure (Andrade et al. 2016), and more recent surveys report significant marine litter accumulation along the coast (Salinas et al. 2024). Environmental stressors also include recurrent harmful algal blooms (HABs), documented since the 1970s, which have periodically impacted fisheries resources (Guzmán and Campodónico 1975, Guzmán et al. 2002). Historical events, such as the 1974 B/T Metula oil spill, highlight the long-term vulnerability of benthic and coastal communities to industrial accidents, compounded at the time by the lack of ecological baseline data (Guzmán 1976, Straughan 1977). In addition, emerging concerns point to the potential impacts of future hydrogen energy developments on the Strait's ecologically and economically valuable habitats (Cruz 2024). Expanding industrial activities may introduce further stressors to an already dynamic marine environment, characterized by strong natural variability driven by oceanographic and climatic processes (Antezana 1999, Valdenegro and Silva 2003, Aracena et al. 2015).

Global studies suggest that the resilience of marine species and ecosystems to anthropogenic pressures may be more limited than previously thought (Malakoff 1997, Hughes et al. 2005, Rombouts et al. 2013). Although the effects of human activities on marine systems are well-documented, accurately quantifying these impacts remains a significant challenge (Vitousek et al. 1997).

In the Magellan region, the limited understanding of invertebrate taxonomy and the scarcity of quantitative biodiversity assessments, as highlighted by Ríos et al. (2003, 2007),

continue to hinder the recognition of biodiversity value in these ecosystems, with most studies remaining isolated and case-specific (Ríos and Mutschke 1999, Cari et al. 2024). Nevertheless, the Strait sustains a diverse benthic and pelagic community shaped by strong spatial gradients. Intertidal and subtidal zones are dominated by mussels *Aulacomya atra*, *Mytilus chilensis* and limpets *Nacella magellanica*, *N. deaurata*, while kelp forests *Macrocystis pyrifera* enhance biodiversity and habitat heterogeneity, supporting invertebrates, fish and mammals (Ríos et al. 2003, 2007, Thatje and Brown 2009). Infaunal assemblages (i.e. benthic fauna living among the sediments) are dominated by polychaetes, which comprise up to 60% of benthic biomass, followed by mollusks with 9–10% (Ríos and Gerdes 1997, Cattaneo-Vietti et al. 1999, Montiel et al. 2011). Furthermore, benthic assemblages in the Strait of Magellan do not form sharply distinct communities but rather a continuum shaped by substrate type, depth, and current velocity, as highlighted by Gutt et al. (1999). Benthic habitats are critical for ecosystem functioning, providing three-dimensional structures and microhabitats that sustain biodiversity, nutrient recycling, and regulation of organic matter (Carvalho et al. 2017, Försterra et al. 2017, Valdivia et al. 2019, Haro et al. 2022, Soukup et al. 2022). Yet, their ecological significance is often overlooked by taxonomic approaches alone. This underscores the urgent need for ecological modeling as a complement to traditional biodiversity assessments. In data-limited systems, food web topology modeling can help identify taxa pivotal to energy flow and network stability (Jordán et al. 2006, Scotti et al. 2017), advancing integrated trophic frameworks that link benthic and pelagic compartments and support ecosystem-based management and conservation.

Recent research has begun to address these gaps through quantitative ecosystem models. Haro et al. (2025) developed the first Ecopath model for the Strait of Magellan, revealing the strong influence of marine mammals on food web dynamics. Killer whales *Orcinus orca* exert top-down control on sea lions and seabirds, indirectly favoring commercially important fish such as the Patagonian blenny *Eleginops maclovinus*. Likewise, humpback whales *Megaptera novaeangliae* contribute to energy transfer between pelagic and benthic compartments by preying on Fuegian sprats *Sprattus fuegensis* and squat lobsters *Grimothea gregaria*.

While biomass-based models have provided valuable insights into the trophic impact of top predators (Pauly 2000), a broader ecological perspective is needed to fully comprehend the structure and function of the Strait of Magellan ecosystem. Beyond the role of apex predators, understanding the diversity of whole trophic compartments is key to evaluating the resilience and stability of the food web (Barnes et al. 2018, Thompson et al. 2020).

Variations in trophic structure and energy flow further emphasize the heterogeneity of this ecosystem. Existing studies have mainly addressed intertidal and shallow subtidal habitats – components of the broader food web of the Strait – showing relatively simple intertidal food webs dominated by grazing gastropods, filter-feeding bivalves, and their predators

(Guzmán and Ríos 1986). Stable isotope analyses in shallow benthic communities have also revealed a high proportion of omnivorous species, suggesting a trophic continuum rather than discrete trophic levels (Gillies et al. 2012, Andrade et al. 2016).

Food webs provide a powerful framework to analyze ecosystem structure and functioning, linking trophic (predator–prey) interactions and energy transfer across trophic levels (Dunne 2005). When food webs comprise more than 25 trophic species or nodes and more than 100 trophic interactions or links, the network approach becomes a suitable analytical tool to characterize the properties of the food web as well as the species' role (Belgrano et al. 2005, Marina et al. 2018b, Delmas et al. 2019). In high-latitude marine environments like the Strait of Magellan, characterized by strong environmental variability and unique ecological dynamics, food web studies are particularly valuable (Rodríguez et al. 2022, Marina et al. 2024, Scian et al. 2025). Moreover, such an approach helps to assess ecosystem resilience – the capacity of an ecosystem to recover its structure and function after disturbance (Pimm 1984, Levin and Lubchenco 2008) – and ecosystem stability, understood as the potential to maintain its structure and function over time in the face of minor perturbations (Grimm et al. 1992, Van Meerbeek et al. 2021). These insights are critical for understanding trophic organization, energy flow, and ecosystem vulnerability to anthropogenic impacts (Mestre et al. 2022).

To further elucidate the trophic structure of the Strait's food web, we applied a topological network approach (presence/absence of interactions), a framework that has proven instrumental in marine ecosystem studies, particularly in high-latitude regions (Marina et al. 2018a, 2024, Rodríguez et al. 2022). This method enables the identification of trophic interactions, energy flow, and potential keystone species, as well as ecosystem responses and vulnerabilities (Dunne et al. 2002a, Belgrano et al. 2005, Marina et al. 2018b, 2024, Rossi et al. 2019).

This research addresses the pressing need to understand the trophic structure and potential response to perturbations of the Strait of Magellan's ecosystem, given its high biodiversity and the diverse array of anthropogenic activities that depend on it. For this, we analyze for the first time the complexity and structure of the food web, and the species' roles by means of a network approach. The findings of this study are intended to guide sustainable management and conservation efforts within this strategic marine territory.

Material and methods

Study area

The Strait of Magellan, one of the few natural waterways connecting the Pacific and Atlantic Oceans (Aldea and Rosenfeld 2011; Fig. 1), extends approximately 570 km in length, with widths varying between 4 and 37 km and depths ranging from 30 to 1100 m (Lutz et al. 2016). Its unique geomorphology, semidiurnal tidal regime, and low salinity,

influenced by precipitation and freshwater inputs, create highly heterogeneous environmental conditions (Brun et al. 2020, Aravena-Yáñez et al. 2025). The Strait supports recreational activities due to its unique natural features, including the presence of marine mammals, as well as established marine routes of international importance, and invaluable cultural heritage, making it a region of rich ecological and socio-economic value (Vergara et al. 2021).

The Strait is a multi-structured system with strong spatial heterogeneity in geomorphology, hydrodynamics and trophic conditions. As described by Cattaneo-Vietti et al. (1999), ecological processes within the system are shaped by pelagic–benthic coupling, driven by variations in primary productivity, the presence of both high- and low-quality particulate organic matter, and intense resuspension processes. These dynamics highlight the dynamic interplay between physical and biological factors that define the ecological significance of the Strait.

Located within the central micro-basin of the Strait, the Paso Ancho – the widest section of the waterway – has been reported to exhibit high phytoplankton standing stocks (Iriarte et al. 2001) and significant primary production (Magazzú et al. 1996), supporting a zooplankton community dominated by copepods, particularly during spring phytoplankton blooms (Hamamé and Antezana 1999). Under favorable conditions, suspension-feeding species such as mussels thrive, contributing to the relatively high benthic secondary production in the Magellan Region, which is predominantly driven by mollusks, polychaetes, and arthropods that strongly mediate benthic energy flow and nutrient cycling (Brey and Gerdes 1999, Cattaneo-Vietti et al. 1999, Ríos and Mutschke 1999).

Food web dataset

We conducted a long-term review over a 50-year period (1975–2025) across the Magellanic region, compiling a detailed dataset from 98 sources. This dataset documents prey–predator interactions by listing the scientific and common names of both predators and prey, along with bibliographic references from journal articles, books, technical reports, theses and local studies. The referenced studies included standardized sampling protocols, field and experimental observations, personal communications, and various analytical approaches including stable isotope and fatty acid analysis.

To compile the dataset, systematic searches were conducted in academic databases (Web of Science, Scopus, Google Scholar, SciELO) and national/international repositories (Biblioteca Digital Aike, <https://bibliotecadigital.umag.cl/>; <https://repositorio.uchile.cl/>; <https://repositorio.unab.cl/>; <https://repositorio.uc.cl/>; <http://opac.pucv.cl/>; <https://bibliotecadigital.exactas.uba.ar/>; <https://aquadocs.org/>; <https://figshare.utas.edu.au/>; <https://www.globalbioticinteraction.org/>; <https://epic.awi.de/>) using keywords such as 'predator–prey interactions,' 'trophic dynamics,' 'Magellanic region,' 'Magellan region,' 'Strait of Magellan,' 'Magellan Strait,' 'feeding behavior,' 'diet,' 'diet composition' and 'food'. Boolean

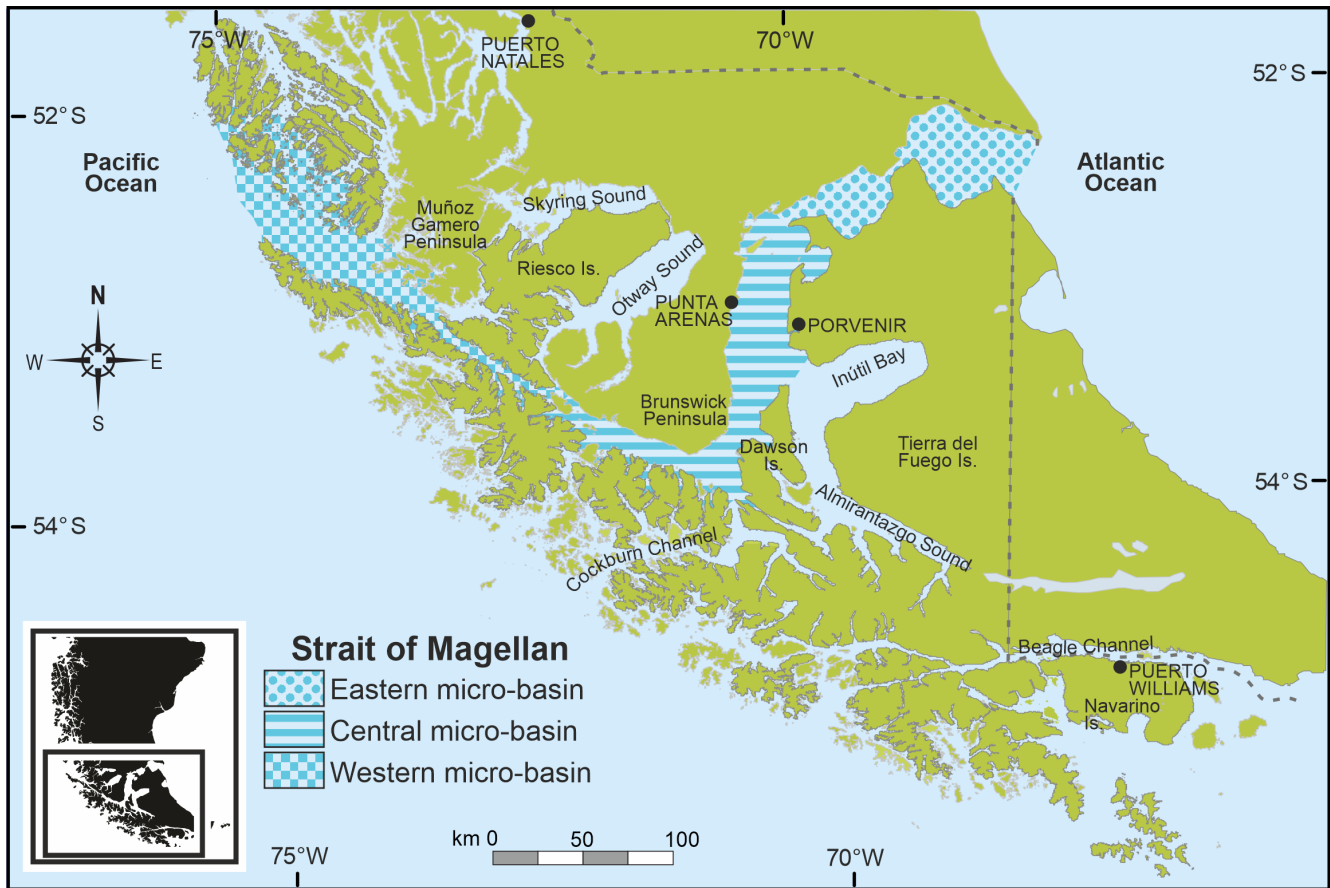


Figure 1. The Strait of Magellan, an important Sub-Antarctic waterway in South America, is the geographical area investigated in this study. The map illustrates the three distinct micro-basins within the Strait: the eastern micro-basin (represented by blue circles), the central micro-basin (represented by blue lines), and the western micro-basin (represented by blue squares).

operators (e.g. AND, OR) were applied to refine results, and inclusion criteria focused on studies that reported specific prey-predator interactions, trophic relationships, or feeding behaviors in marine ecosystems. Articles were screened for relevance, and only studies meeting these criteria were incorporated into the dataset. The keyword search was conducted in both Spanish and English to ensure a comprehensive collection of relevant studies and sources, broadening the dataset to include local, regional and international literature.

The trophic interactions included in this analysis represent studies conducted across the western, central, and eastern micro-basins of the Strait, as identified in previous oceanographic and ecological assessments (Valdenegro and Silva 2003). This integrative approach allows for a spatially comprehensive representation of species interactions, accounting for biogeographic and environmental variability along the longitudinal axis of the Strait. The list of interactions that comprise the food web of the Strait of Magellan is in the Supporting information.

A trophic species is defined as a group of taxa that share the same set of predators and prey within a food web. This grouping is functional rather than taxonomic, and is commonly used in food web models without losing the functional

structure of the system (May 1973, Gauzens et al. 2013). Hereafter, a ‘trophic species’ will denote any food web’s node, regardless of its taxonomic resolution (e.g. species, genus, family).

Food web analysis

We analyzed the topological food web of the Strait of Magellan by means of widely-used network properties describing the complexity and structure in food web studies (Dunne et al. 2002a, Landi et al. 2018, Marina et al. 2024). The properties that we used were: connectance, degree distribution, generality, vulnerability, omnivory and small-world pattern (Table 1). Regarding the degree distribution or the distribution of interactions among species, we calculated the in-degree distribution, which represents the number of prey per predator (generality), the out-degree distribution, which represents the number of predators per prey (vulnerability), and the total-degree distribution, commonly named degree distribution, which accounts for all species’ interactions (in + out). For the latter, we performed a model fit using the cumulative distribution and considering the following models: power law, exponential, log-power law, and log-exponential. The Akaike information criterion (AIC) and Bayesian information

Table 1. Food web properties at the network and node (species) level, analyzed in this study. For each property a definition and ecological interpretation is provided.

Property	Symbol	Definition and ecological interpretation	Reference
Network level			
Number of species	S	Total number of trophic species (nodes) in a food web.	May 1973, Tilman 1996
Number of interactions	L	Total number of prey–predator or trophic interactions (links) in a food web. It represents the number of pathways along which energy flows in an ecosystem.	Dunne et al. 2002a
Density of interactions	L/S	Mean number of interactions per species. It informs how connected species are in a food web.	Dunne et al. 2002a
Connectance	C	Proportion of established trophic interactions among species (L) relative to all possible ones (S^2). It represents a fundamental measure of food web complexity. Estimator of food web robustness to perturbations.	May 1973, Dunne et al. 2002a, Montoya et al. 2006
Degree distribution	DD	Frequency of trophic species with k or more interactions in a food web. Estimator of food web vulnerability to random failures and intentional attacks (species extinction).	Albert and Barabási 2002
Generality	Gen	Frequency of predators with g or more prey in a food web. It represents the number of prey of a predator and its diet breadth. Estimator of generalist/specialist predators in a food web.	Schoener 1989
Vulnerability	Vul	Frequency of prey with v or more predators in a food web. It represents the number of predators of a prey. Estimator of most/least demanded prey in a food web.	Schoener 1989
Omnivory	Omn	Proportion of predators that feed on prey at different trophic levels. Estimator of trophic flexibility of an ecosystem with implications on food web stability depending on the interaction strength distribution.	McCann and Hastings 1997
Small-world pattern	SW	Compared to random networks, this network exhibits a relatively short mean path length (short distance between nodes) and a high clustering coefficient (strong tendency to form tightly interconnected compartments). The implications of this structural pattern in food webs are crucial for understanding evolutionary trajectories and assessing vulnerability to disturbances.	Watts and Strogatz 1998, Montoya and Solé 2002
Node level			
Trophic position	TP	Position of a species relative to the primary source of energy in a food web (basal species). It represents the number of interactions separating an organism from the base of production, and indicates basal, intermediate and top species.	Lindeman 1942, Thompson et al. 2007
Degree	Deg	Total number of interactions of a species, considering in- (number of predators) and out-interactions (number of prey). Species with a high degree are the most connected, so their influence on the robustness of the network is high.	Delmas et al. 2019
Closeness	Clo	Distance of a species to every other species in the food web, calculated as the inverse of the sum of distances (number of interactions). Estimator of how efficiently perturbations on a species are likely to influence the food web.	Freeman 1978, Delmas et al. 2019
Betweenness	Btw	Frequency of a species in the shortest path between a pair of species. It measures the influence of species loss in fragmentation processes and the potential dispersion of disturbances.	Freeman 1978, Delmas et al. 2019
Topological role	TR	Role of a species according to the proportion of interactions within and across modules. Four roles are defined: network connector, module connector, module hub and module specialist. Network and module connectors maintain the connectivity of the food web.	Guimerà and Nunes Amaral 2005, Kortsch et al. 2015

criterion (BIC) were used as the model selection criteria. Moreover, with the aim of exploring the distribution of interactions among functional groups we grouped species in 38 taxa, the majority at the Class or Order level. For clarity, we decided to graph and report only those groups with more than three species: Macroalgae; Non-living (Detritus, Necromass); Copepoda; Bivalvia; Gastropoda; Echinoidea; Decapoda; Asteroidea; Teleostei; Aves; Cephalopoda; Mammalia. To test for the small-world pattern, we compared the empirical clustering coefficient and path length of the food web against a null distribution. This distribution was generated from 1000

random networks using the Erdős–Rényi model (Erdős and Rényi 1959), which were constrained to the same number of species (S) and interactions (L) as our empirical web, following the methodology of Marina et al. (2018b).

These network properties were selected to better understand how the food web may respond to ongoing disturbances in the Strait, including marine pollution, industrial development, and the still emerging invasion of salmonids resulting from escapes from salmon farms (Correa and Gross 2008, Da Silva et al. 2023, Giesecke et al. 2024, Salinas et al. 2024), an issue understudied at the ecosystem level in the

Strait of Magellan. Table 1 provides a definition and ecological interpretation for each of the mentioned network-level properties.

We also performed analyses at the node level (Delmas et al. 2019), with the aim of characterizing species' roles. The properties that we used were: trophic position, degree, closeness, betweenness and topological role (Table 1). Topological roles are a by-product of modularity, a property at the network level, that describes the tendency of food webs to organize into non-random, modular structures, where modules consist of species that interact more frequently with each other than with species outside the module (Guimerà and Nunes Amaral 2005). We used the cluster spinglass algorithm, performing 1000 simulations, to partition the food web into modules that maximize the modularity. This algorithm identifies significant community structure by rewarding partitions where the density of interactions within modules is greater than would be expected by chance. We determined the topological role of each species using the functional cartography approach (Guimerà and Nunes Amaral 2005). This method classifies nodes based on two key parameters: 1) the within-module degree (dz_i), a z-score that measures a species' i connectivity within its own module, and 2) the participation coefficient (PC_i), which quantifies how a species' i interactions are distributed among different modules. Following Kortsch et al. (2015), we then assigned a specific role to each species by partitioning the $dz - P$ parameter space into four regions using threshold values of $dz = 2.5$ and $PC = 0.625$. Specifically, we classified each species into one of four roles: 'module hub' ($dz \geq 2.5$, $PC < 0.625$), characterized by a high number of interactions concentrated within its own module; 'module specialist' ($dz < 2.5$, $PC < 0.625$), having relatively few interactions mainly within its module; 'module connector' ($dz < 2.5$, $PC \geq 0.625$), with limited interactions primarily occurring between modules; and 'network connector' ($dz \geq 2.5$, $PC \geq 0.625$), exhibiting high connectivity both within and across modules. See Table 1 for definition and ecological meaning of the other node-level properties (trophic position, degree, closeness and betweenness).

Based on selected node-level properties, we created the Keystone Species Index (hereafter KSI). This index considers the so-called centrality indices: degree, closeness and betweenness. To do this, we ranked each species according to each of the centrality indices: the higher the degree, closeness and betweenness, the higher the species rank. Then we averaged these three rankings to obtain the KSI for each species. When two or more species had the same KSI value, they shared the same position in the ranking. The KSI value of a species is inversely related to its centrality ranking. A KSI of 1 indicates the highest-ranked (most central) species, while a value equal to the total number of species indicates the lowest-ranked (least central) species. Moreover, we considered the trophic position and topological role to complement the results of the KSI and comprehend the species' role.

All analyses were performed using the R ver. 4.4.1 (www.r-project.org), particularly packages 'igraph' ver. 2.1.2 (Csárdi et al. 2025), 'NetIndices' ver. 1.4.4.1 (Kones et al.

2009), 'NetworkExtinction' ver. 1.0.3 (Ávila-Thieme et al. 2023), and 'multiweb' ver. 0.6.9 (Saravia 2025). The following are the main R functions used to perform the analyses: degree distribution (NetworkExtinction::DegreeDistribution()), small-world (multiweb::calc_swness_zscore()), topological role (multiweb::calc_topological_roles()) and multiweb::classify_topological_roles()). The complete source code and data are available at <https://zenodo.org/records/17344507>.

Results

Food web properties

The food web of the Strait of Magellan (Fig. 2) is composed of 139 trophic species (S), where 3 are non-living nodes (sediment, detritus and phytodetritus). Notably, 78.42% of nodes are taxonomically resolved at the Genus or Species level. The network complexity of the food web includes a total of 435 (L) interactions, a density of interactions (L/S) of 3.13 and a connectance value (L/S²) of 0.022. More than half of the species (56.12%) occupy intermediate trophic positions, meaning they are involved in interactions as both prey and predators. Intermediate-level species outnumber top (33.09%) and basal species (10.79%). Additionally, omnivory accounts for 50.36% of the total species in the food web. The mean trophic level of the network is 2.77, and the top predator with the maximum trophic position is 5.57 (killer whale or *Orcinus orca*).

Moreover, path length and clustering coefficient were calculated: 1.88 and 0.09, respectively. The food web presents a small-world pattern (Supporting information). The

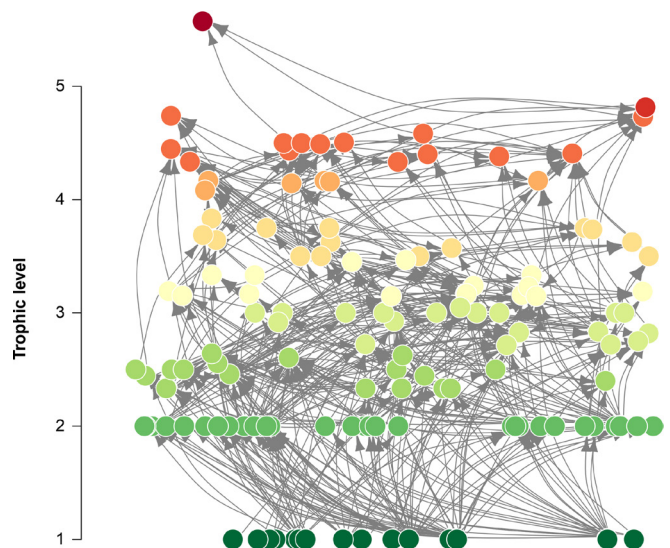


Figure 2. Graph of the food web of the Strait of Magellan. Circles represent trophic species and arrows represent predator-prey interactions, where the direction depicts the flow of energy in the ecosystem. Vertical position and color gradient indicate trophic level (max. trophic position = 5.57).

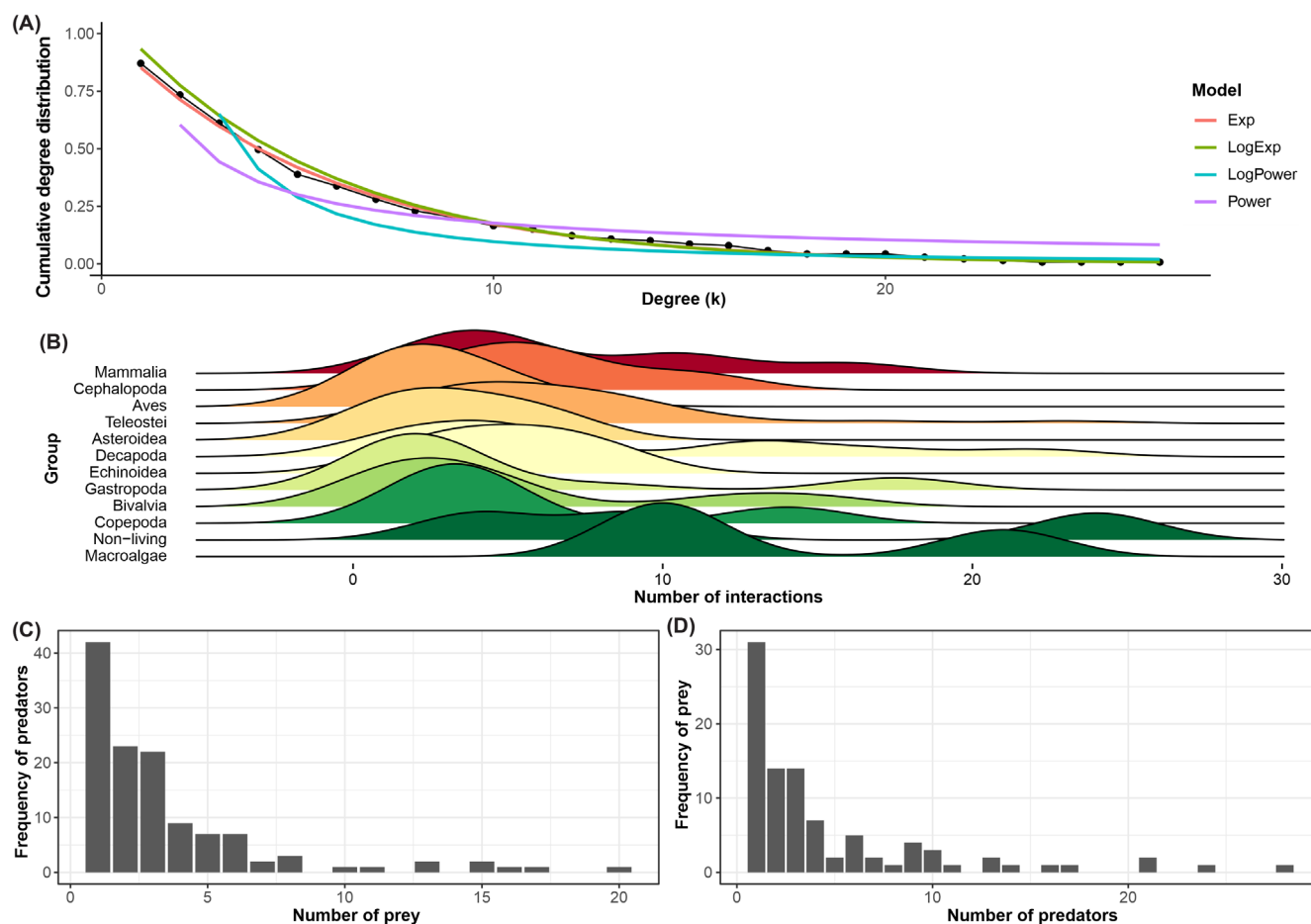


Figure 3. Distribution of predator–prey interactions in the food web of the Strait of Magellan. (A) Cumulative. Best fit model is exponential. (B) By group. Groups are vertically ordered by increasing trophic level (following coloration of Fig. 2); groups with fewer than 3 species were not plotted (e.g. Foraminifera, Ostracoda). All groups and the species that comprise them are shown in the Supporting information. (C) Generality or distribution of prey among predators. (D) Vulnerability or distribution of predators among prey. Fitting details for cumulative distribution are provided in the Supporting information.

degree distribution of the food web fits an exponential model (Fig. 3A, Supporting information).

Species properties

When interactions were analysed by group (Fig. 3B), most were concentrated in a limited number of species. The groups with the highest number of interactions, each comprising more than one species, included: the Patagonian blenny and Fuegian sprat (Teleostei); the limpets *Nacella deaurata* and *N. magellanica* (Gastropoda); the southern king crab *Lithodes santolla* and the false king crab *Paralomis granulosa* (Decapoda); the South American sea lion *Otaria byronia* (Mammalia); and *Mytilus* spp. (Bivalvia). Overall, the distribution of interactions across the network shows a clear asymmetry among different organizational levels, structured by taxonomic or functional groups, as well as by prey and predator roles.

The three nodes with the highest degree were Phytoplankton, Detritus and the Patagonian blenny, with 28, 24 and 23 interactions, respectively. The distributions of generality (number of prey per predator) and vulnerability

(number of predators per prey) differed slightly in shape (Fig. 3C–D). Most predators have few prey species, with a mean of 3.51 prey per predator. In contrast, vulnerability is more asymmetric, indicating that while most prey are consumed by only a few predators, a small number of prey species are highly demanded (with a mean of 4.68 predators per prey). This reflects an uneven structure in the way prey and predators are distributed across the network.

The role of species within the network varies according to their trophic position. In terms of degree, the most connected species were those at the basal levels (Phytoplankton and Detritus), as reflected by larger circle sizes (Fig. 4). For closeness, species at intermediate trophic levels (TP=2.5–3.5), such as the Patagonian blenny, Copepoda and Polychaeta, emerged as the most important (Fig. 5). Similarly, betweenness was highest for species at these intermediate trophic levels, including Polychaeta, the Patagonian blenny and the Fuegian sprat (Fig. 6).

According to the Keystone Species Index (KSI), the top 10 most influential trophic species in the Strait of Magellan's

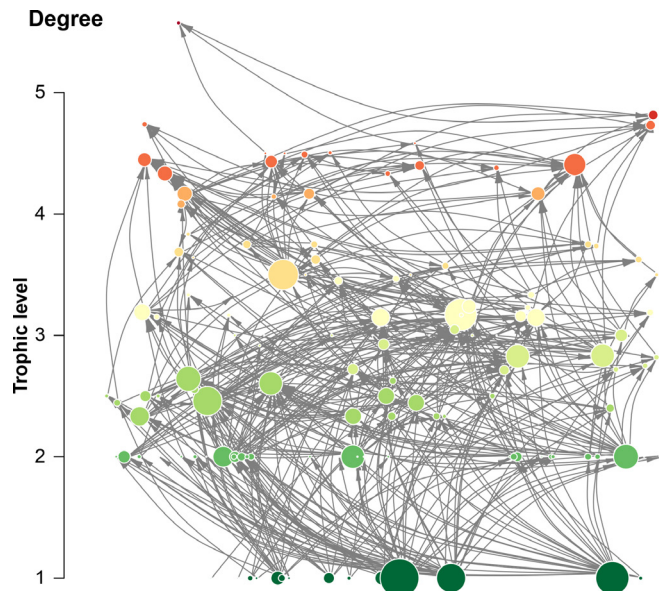


Figure 4. Degree for the species of the food web of the Strait of Magellan. Circle size is relative to the importance of the corresponding property. Circles represent trophic species and arrows represent predator–prey interactions, where the direction depicts the flow of energy in the ecosystem. Vertical position and color indicate trophic level. Refer to the Supporting information for node-level properties of species.

food web include the Patagonian blenny, Polychaeta, Benthic Decapoda, Copepoda, the limpet *Nacella deaurata*, the Fuegian sprat, Amphipoda, the southern king crab, the rockcod *Patagonotothen tessellata*, Foraminifera and Ostracoda (Table 2).

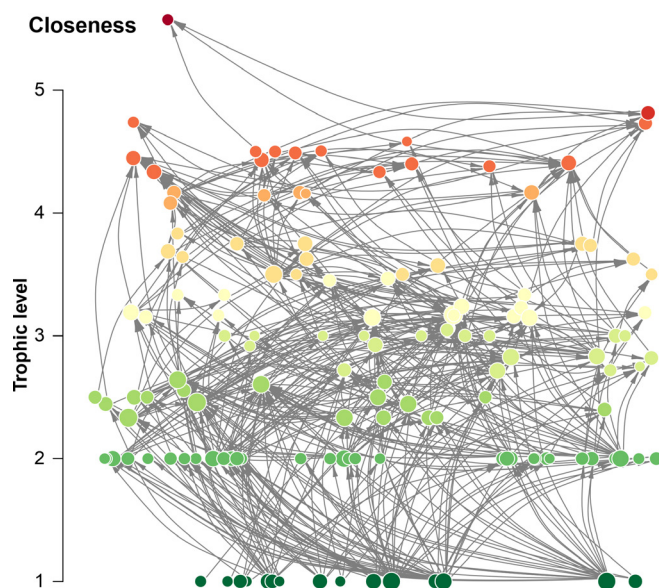


Figure 5. Closeness for the species of the food web of the Strait of Magellan. Circle representation, size and vertical position indicate the same as Fig. 4. Refer to the Supporting information for node-level properties of species.

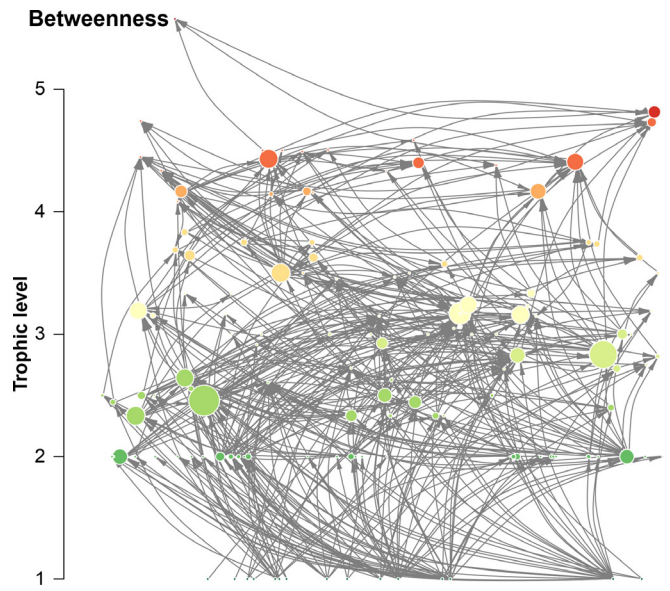


Figure 6. Betweenness for the species of the food web of the Strait of Magellan. Circle representation, size and vertical position indicate the same as Fig. 4. Refer to the Supporting information for node-level properties of species.

When analyzing the topological role of the 139 trophic species within the food web, we identified 120 species as module specialists (i.e. species with few links primarily within their own module), 15 as module connectors (i.e. species with few links mainly between modules), 2 as module hubs (i.e. species with a high number of links mostly within their module), and 2 as network connectors (i.e. species with high connectivity both within and between modules) (Fig. 7). Key species identified by the KSI were also recognized as module connectors, including teleost fishes such as the notothenioids Patagonian blenny and the rockcod. Other connector species included the notothenioid *Patagonotothen sima* (KSI ranking=25), as well as the squat lobster (KSI ranking=28). In general, these module connectors span diverse taxonomic groups, including Ostracoda, Polychaeta, Cephalopoda, Copepoda, Gastropoda, Bryozoa, and Ophiuroidea. Additionally, lower-trophic level species, such as Phytodetritus and Benthic diatoms, played a similar role as module connectors. Moreover, detritus and phytoplankton emerged as network connectors. Finally, Brown macroalgae and Benthic Decapoda functioned as the module hubs within the Strait's food web.

Discussion

Food web properties

Understanding how food web structure shapes ecosystem functioning is fundamental to predicting responses to disturbances (Dunne 2005, Montoya et al. 2006, Martins et al. 2024). In this study, we examined key network properties of the food web in the Strait of Magellan to explore patterns of complexity, structure and potential vulnerability to stressors. Our findings contribute to ongoing efforts to understand how

Table 2. Keystone Species Index (KSI) for the species of the Strait of Magellan food web. The rank accounts for the total number of interactions (degree), the species' role as a bridge between other species (betweenness), and its proximity to all other species within the network (closeness). Node-level properties are also presented, including degree, betweenness, closeness and trophic position (TP).

Trophic species/taxa	Degree	Closeness	Betweenness	KSI	TP	Rank
<i>Eleginops maclovinus</i>	23	0.0034	60.83	2.33	3.17	1
Polychaeta	21	0.0033	120.84	2.67	2.46	2
Benthic Decapoda	22	0.0032	46.22	4.33	3.50	3
Copepoda	14	0.0034	45.37	5.67	2.30	4
<i>Nacella deaurata</i>	18	0.0032	40.98	6.33	2.64	5
<i>Sprattus fuegensis</i>	17	0.0030	98.14	7.67	2.83	6
Amphipoda	18	0.0031	26.35	10.00	2.00	7
<i>Lithodes santolla</i>	17	0.0030	29.53	11.00	2.83	8
<i>Patagonotothen tessellata</i>	12	0.0029	38.62	13.00	3.19	9
Foraminifera	12	0.0031	21.02	13.00	2.44	9
Ostracoda	12	0.0031	17.90	13.33	2.33	10

the network of trophic interactions reflects and potentially mediates ecosystem responses, particularly in high-latitude marine systems (Rodríguez et al. 2022, Marina et al. 2024).

Our results reveal a relatively low connectance for the Strait of Magellan food web ($C=0.022$), a value at the lower end of the range reported for adjacent Sub-Antarctic marine ecosystems (0.01–0.05; López-López et al. 2022, Rodríguez et al. 2022, Marina et al. 2024, Scian et al. 2025). Notably, the Beagle Channel exhibits the highest connectance within this range ($C=0.05$; Rodríguez et al. 2022), highlighting the influence of local ecological drivers on food web structure even across geographically proximate systems (Saravia et al. 2022). Connectance is a good proxy for stability in topological food webs, as low-connected networks (< 0.05) are highly sensitive to the loss of highly-connected nodes, which

can reduce robustness, lower resistance to invasions, and trigger cascading extinctions (Dunne et al. 2002a, Estrada 2007, López-López et al. 2022). Indeed, the removal of as little as one-quarter of species can cause network collapse (Dunne et al. 2002b). By contrast, more connected food webs are generally more resilient to disturbances such as invasions and local extinctions (Dunne et al. 2002b, Montoya and Solé 2002, Smith-Ramesh et al. 2017), underscoring the buffering role of connectance in maintaining ecosystem stability.

Furthermore, such a low connectance combined with an asymmetric degree distribution (exponential fit) and small-world pattern observed in the Strait's food web suggests increased susceptibility to perturbations, particularly in the event of the loss of highly connected species or the alteration of interactions through biological invasions (Marina et al.

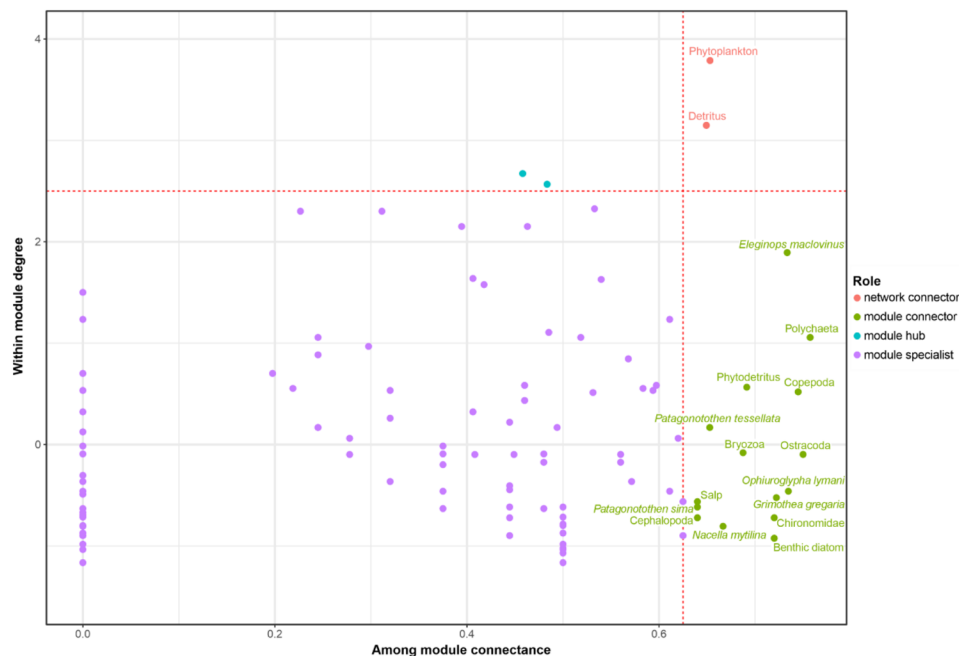


Figure 7. Topological role for the species of the food web of the Strait of Magellan. Four roles, according to 'within module degree' and 'among module connectance' parameters, are possible: network connector, module connector, module hub and module specialist. Each colored dot represents a trophic species and its topological role. Labelled dots correspond to network connectors and module connectors. Refer to the Supporting information for the topological role of each species.

2018b, Rodriguez et al. 2022, Jordán et al. 2024). Such topological structure may reflect a combination of trophic specialization, environmental filtering, and other ecological constraints shown in marine ecosystems (Link 2002, Kortsch et al. 2019, Mestre et al. 2022). Similar patterns have been described in other high-latitude marine ecosystems, where low connectance coexists with moderate to high levels of omnivory (Rodriguez et al. 2022). This suggests that many species feed across multiple trophic levels but on relatively few prey species – an omnivory pattern that reflects vertically dispersed yet narrow interactions. In particular, habitat heterogeneity and microhabitat segregation – consistent with the spatial patchiness described for the Strait of Magellan (Arntz 1999, Gutt et al. 1999, Ríos 2007) – may constrain the number of realized trophic interactions, even when omnivory is present.

Stable isotope studies support this interpretation, describing a trophic continuum in benthic communities where energy flows across levels but largely from a few basal sources (Andrade et al. 2016). Together, these results suggest a vertically integrated yet functionally constrained system, where energy transfer is concentrated along dominant pathways, making the network potentially vulnerable to perturbations affecting key prey or basal resources.

Moreover, research on functional diversity in the Magellan Region – such as in Parry Bay, Almirantazgo Sound, a proglacial fjord influenced by glacial meltwater – reveals high functional redundancy among benthic taxa, likely shaped by strong environmental filtering (Sepúlveda et al. 2024). In contrast, the more stable and heterogeneous conditions of the Strait of Magellan could support a broader array of traits, enabling greater trophic flexibility. Thus, although our analysis shows low connectance, the presence of omnivory and trophic niche overlap in some compartments may reflect a partial trophic continuum, as previously suggested for benthic communities in the region (Andrade et al. 2016).

The food web of the Strait of Magellan also shows asymmetry in both the overall distribution of interactions and in the distribution of prey among predators (generality), and predators among prey (vulnerability). This pattern aligns with previous findings from Sub-Antarctic regions such as Burdwood Bank and Yaganes (Rodriguez et al. 2022, Marina et al. 2024, Scian et al. 2025). While networks with exponential degree distributions are typically prone to collapse, studies on high-latitude marine ecosystems have shown resilience to species loss without complete collapse (Cordone et al. 2018, 2020). Disturbances to generalist species are more likely to trigger cascading extinctions, while disturbances to specialist species have less severe effects on network stability (McCann and Hastings 1997, Rodriguez et al. 2022). In the Strait of Magellan, however, the most highly connected species – such as the Patagonian blenny and polychaetes – might be at risk due to external pressures, which include unregulated fishing pressure (Guzmán and Campodónico 1973), while for polychaetes, their environmental sensitivity makes them vulnerable to climate-driven stressors (Riedel et al. 2012, Herrera-Perez and Méndez 2019), issues further discussed in the next section.

Furthermore, the food web of the Strait displays a high degree of clustering and short path lengths between species, features that are characteristic of a small-world architecture (Watts and Strogatz 1998, Montoya and Solé 2002). These properties are commonly associated with complex food webs presenting multiple, redundant energy pathways (Eskuche-Keith et al. 2023). In particular, this architecture facilitates the rapid spread of perturbations, such as biodiversity loss, marine contamination, or overfishing, throughout the network (Delmas et al. 2019). However, the high clustering observed in this food web suggests the formation of tightly connected subnetworks, which can localize disturbances and prevent their widespread impact (Kortsch et al. 2019, Heer et al. 2020). This balance between connectivity and compartmentalization enhances the adaptive capacity of the ecosystem, allowing for a more efficient response to environmental fluctuations (Montoya and Solé 2002, Stouffer and Bascompte 2011, Gilarranz et al. 2017). Complementing these topological features, our findings of widespread omnivory – exceeding 50% of trophic species – further underscore the mechanisms that enhance food web resilience. Omnivory promotes trophic redundancy and dietary flexibility (Fagan 1997, Borrvall et al. 2000, Neutel et al. 2002), allowing the network to better buffer fluctuations in prey availability and thereby enhancing its persistence (Stouffer and Bascompte 2011). In the Strait, many species occupy intermediate trophic levels, functioning simultaneously as predators and prey, a structural trait that facilitates energy flow and reinforces connectivity across the network. Although this proportion is slightly lower than in other sub-Antarctic marine ecosystems (e.g. 67% Beagle Channel, Rodriguez et al. 2022; 71% Yaganes, Scian et al. 2025), it remains a key attribute. Moreover, higher taxonomic resolution may reveal an even greater share of intermediate species. The prevalence of omnivory and intermediate positions has not only been previously reported for the Strait (Andrade et al. 2016), but also appears as a recurrent feature in other high-latitude systems (Rodriguez et al. 2022, Marina et al. 2024, Scian et al. 2025), likely reflecting the influence of environmental variability and resource diversity in promoting generalist trophic strategies. Taken together, our findings suggest that the Strait of Magellan's food web exhibits a combination of properties associated with both robustness and vulnerability. Intermediate consumers and widespread omnivory may enhance the system's capacity to buffer against certain disturbances. However, the asymmetric distribution of interactions and the dependence on a few highly-connected species could constrain the system's ability to absorb more severe or prolonged stressors.

Species' role in the Strait of Magellan food web

Classical ecological theory suggests that keystone species play a fundamental role in structuring ecosystems and shaping food web dynamics by regulating species abundance and driving critical ecological processes (Paine 1969, Piraino et al. 2002). However, the definition of keystone species remains debated, as there is no universally accepted criterion for their identification, especially in heterogeneous and variable systems

(Cottee-Jones and Whittaker 2012, Fulton and Sainsbury 2024, Jordán et al. 2024).

The few biomass-based modeling approaches applied in the Strait of Magellan such as that of Haro et al. (2025) have identified apex predators as keystone species, reflecting a traditional view that emphasizes top-down control in ecosystem analysis. In contrast, our analysis incorporates species from multiple trophic levels – ranging from primary producers and detritus to top predators – and explicitly disaggregates the benthic compartment, which is often treated as a single, generalized component. This enables a more inclusive perspective in which mid-trophic level species can also play important roles by acting as trophic intermediaries, linking multiple species and, ultimately, contributing to overall food web stability (Jordán 2009, Scian et al. 2025).

In this sense, our network analysis has identified key species with the highest potential to maintain the trophic structure of the Strait of Magellan ecosystem. These species occupy mid-trophic positions in the network ($TP = 2.5\text{--}3.5$), characterized by predominantly omnivorous diets and some functioning as module connectors.

Among them, the Patagonian blenny ($KSI = 2.33$) inhabits shallow coastal areas and estuaries, and feeds on diverse benthic organisms (Guzmán and Campodónico 1973, Licandeo et al. 2006, Rumbold et al. 2024). In our study, the Patagonian blenny presents a trophic position of 3.17, which aligns with isotopic evidence from the Beagle Channel, where Ricciardelli et al. (2020) reported a similar trophic position for the species ($TP = 3.4$). Together, these findings indicate a consistent intermediate predatory role across Sub-Antarctic ecosystems, reinforcing the species' importance as a connector between benthic and pelagic compartments. Beyond its trophic position, the Patagonian blenny exhibited a module connector role, meaning high trophic connectivity with feeding links that span both benthic and pelagic compartments. This connectivity identifies the species as a functional hub that integrates energy pathways across modules, enhancing network coherence and potentially buffering or transmitting perturbations through the system. Such versatility underscores its capacity to modulate ecosystem dynamics and amplify its influence on community structure by simultaneously shaping prey availability and predator interactions. The ecological importance revealed here contrasts with the species' population decline in recent decades, likely driven by overfishing, predation by introduced salmonids, and biological traits such as hermaphroditism, which increase its vulnerability to size-selective fishing (Guzmán and Campodónico 1973, Pequeño and Olivera 2005, Rumbold et al. 2024). As a connector in the food web, the blenny's decline could trigger cascading effects, impacting both top predators (Chilean and Peale's dolphins; Brickley et al. 2005, Martin and Bastida 2008) and benthic communities it helps regulate through opportunistic feeding (Licandeo et al. 2006). Despite being historically lightly fished (~ 151 tons year⁻¹; Guzmán and Campodónico 1973), its fishery remains unregulated, with no stock assessments conducted to date.

Polychaetes were also identified as the second-ranked species by the KSI ($KSI = 2.67$) and module connectors. Their

high diversity and broad distribution, from intertidal to deep-sea habitats, make them essential in both the soft and hard-bottom communities of the Magellan region, where they represent $\sim 20\%$ of the benthic fauna (Montiel 2005). Despite their ecological relevance, polychaetes may be increasingly vulnerable to climate change-related stressors such as rising temperatures, oxygen depletion, and changes in sediment dynamics (Riedel et al. 2012, Herrera-Perez and Méndez 2019). These environmental changes can particularly affect polychaetes due to their diverse feeding strategies, including suspension and deposit feeding (Fauchald and Jumars 1979), which are tightly linked to water column properties and sediment quality. Alterations in particulate organic matter availability, turbidity, and sedimentation rates could impair their feeding efficiency, survival, and overall functional roles in benthic ecosystems. Moreover, shifts in primary production may reduce the quantity and quality of organic matter reaching the benthos, potentially compromising food availability for detritivorous- and suspension-feeding species (Ríos and Mutschke 1999, Ríos et al. 2003, Hüne and Rivera 2010, Andrade et al. 2022). Polychaetes function as primary or secondary consumers (Bergamino et al. 2011, Jumars et al. 2015), and are preyed upon by notothenioid fishes, southern king crab, and squat lobster (Karas et al. 2007, Hüne and Rivera 2010, Andrade et al. 2022), which suggests an important role in supporting ecosystem productivity and resilience (Brey and Gerdes 1999).

Similarly, Benthic Decapoda act as a key node in the Strait of Magellan ($KSI = 4.33$). Along marine food webs, decapods often dominate invertebrate assemblages and contribute substantially to biomass, secondary production and fisheries, thereby shaping benthic communities and sediment processes (Boudreau and Worm 2012). Presently, both anomuran and brachyuran decapods are among the dominant faunal elements in the Magellan region, which acted as a glacial refuge and has consequently retained a comparatively higher number of species and greater decapod diversity than Antarctic marine ecosystems (Arntz 1999). Nevertheless, this should be viewed as an initial approximation of their relevance, as the low taxonomic resolution of decapods in diet studies based on stomach content analysis remains a challenge for identifying which specific species in the region bear this keystone role. Notably, the squat lobster *Grimothea gregaria*, recognized for its importance in Sub-Antarctic food webs through its abundances, omnivorous diet and vertical mobility that connect benthic and pelagic realms (Romero et al. 2004, 2006, Vinuesa and Varisco 2007, Haro et al. 2016, 2022), did not appear among the top-ranked species in our analysis ($KSI = 29.33$), but did emerge as a module connector, suggesting a relevant role for maintaining the coherence of the food web.

Additional keystone trophic species contributors to the Strait of Magellan food web included copepods, the limpet *Nacella deaurata*, the Fuegian sprat *Sprattus fuegensis*, the southern king crab *Lithodes santolla*, and the rockcod, all of which mediate multiple energy pathways and connect benthic and pelagic compartments. Together, these taxa span different functional groups and habitats, illustrating how a small

number of versatile species underpin network coherence and resilience despite the overall sparsity of connections. At the benthic level, the limpet *N. deaurata* play a dual role of an omnivorous grazer – both consumer of algae and invertebrates and prey for higher predators – emphasizing the importance of shoreline grazers in linking shallow rocky-shore habitats to the wider network (Castilla 1985, Andrade and Brey 2014, Rosenfeld et al. 2018). Among fishes, the Fuegian sprat stands out as one of the most abundant fishes species in the region (Diez et al. 2018, Friedlander et al. 2018), functioning as a key energy conduit between lower and upper trophic levels. It supports a broad range of predators – including penguins, seabirds, humpback whales, sea lions and hakes – interacting with at least 10 functional groups (Haro et al. 2022). Previous studies have identified the Fuegian sprat as a wasp-waist species in Sub-Antarctic marine ecosystems such as the Southwest Atlantic, the Beagle Channel, Tierra del Fuego's Atlantic coast and Burdwood Bank (Riccialdelli et al. 2020). Wasp-waist species are characterized by their high abundance and intermediate trophic position, exerting both bottom-up control on predators and top-down control on lower trophic levels, thereby acting as central nodes in the food web (Hunt and McKinnell 2006). The potential consequences of a decline in Fuegian sprat populations underscore the importance of integrating functional traits into conservation efforts. Despite its ecological significance, however, there is a notable lack of current data on its population trends or fisheries in the Magellan region, emphasizing the need for targeted monitoring and stock assessments.

In addition, our analysis revealed detritus as a network connector. This underscores the central integrative function of detritus in connecting disparate trophic groups and enhancing overall food web cohesion. The prominence of detritus likely reflects the importance of benthic–pelagic coupling in the Strait of Magellan (Cattaneo-Vietti et al. 1999) and may buffer the system against perturbations. When compared with empirical evidence from a stable isotope-based study in the region (Andrade et al. 2016), which identified macroalgae and sediments as key basal carbon sources supporting multiple consumer compartments, our findings provide a complementary structural perspective. Stable isotope analyses reveal the assimilated sources of energy flow (Andrade et al. 2016), while the network-based approach identifies the organizational role of species – such as detritus – in maintaining system connectivity across trophic levels, including top predators. Together, these lines of evidence suggest that prevailing hydrodynamic conditions may facilitate detrital pathways via macroalgal decomposition, reinforcing the ecological centrality of detritus in supporting the resilience and integrity of the Strait of Magellan food web.

Overall, our analysis of species' topological roles in the Strait of Magellan food web reveals a structure dominated by module connectors and specialists, with only two trophic species – both basal sources – functioning as network connectors (i.e. species with numerous links spanning across modules). While highly connected species are generally expected to exert a stronger influence on ecosystem dynamics (Wootton

and Stouffer 2016), the marked asymmetry in interaction distribution observed in this network suggests that a limited subset of species plays a disproportionately critical role in sustaining overall cohesion. These findings underscore the importance of identifying and protecting structurally pivotal species and compartments that underpin the resilience and integrity of the broader ecosystem. The potential loss or decline of such key nodes could significantly disrupt energy transfer efficiency and compromise network stability (Jordán 2009, Lai et al. 2012, Gilarranz et al. 2017), particularly in a system already characterized by low connectance (Dunne et al. 2002a).

Inference on ecosystem response to stressors

This study contributes to ongoing efforts reinforcing the notion that species should not be viewed as isolated elements, but rather as integral components within complex and dynamic trophic systems. Our analysis highlights how ecosystem resilience is supported by food web complexity and structure features and the presence of species acting as structural connectors across multiple trophic levels, both of which contribute to food web cohesion and stability. Within this framework, the identification of keystone species further underscores the importance of functional roles. In the Strait of Magellan, native notothenioid fishes such as the Patagonian blenny and the rockcod exemplify these roles by bridging energy pathways across different trophic compartments. Their central positions in the web may enhance energy flow and contribute to the overall resilience of the system – provided that these structural features remain intact in the face of increasing anthropogenic stressors.

One of the most immediate threats to ecosystem stability in the Strait of Magellan is the southward expansion of aquaculture and the establishment of non-native salmonids (Buschmann et al. 2023, Soto et al. 2023). In southern Chile's freshwater systems, salmonid introductions have already altered native food webs through competition, predation and habitat displacement (Soto et al. 2006, Elgueta et al. 2013, Arismendi et al. 2014, Ortiz-Sandoval et al. 2017). Although no marine or estuarine studies have documented direct interactions with native notothenioid fishes, anecdotal reports suggest such interactions may occur in nearshore habitats (Andrade unpubl.). If similar invasions expand in the Strait, keystone species such as the Patagonian blenny and rockcod could face strong pressures, with cascading consequences for food web structure.

In our network, the two salmonid taxa identified – *Salmo salar* and *Oncorhynchus* sp. – occupy high trophic positions (4.1 and 4.5; Supporting information). While node-level properties indicate low degree and betweenness, their elevated trophic positions imply potential top-down control on prey populations, with impacts propagating through trophic cascades. This interpretation is limited by the absence of prey abundance and density data, which may underestimate their influence. Future studies should quantify these dynamics to better assess salmonid effects in Sub-Antarctic marine ecosystems.

Furthermore, environmental disturbances such as climate change and pollution can strongly affect food web dynamics (Canuel et al. 2012, Campanyà-Llovet et al. 2017). At the same time, microplastic contamination is emerging as a growing concern: in central Chile, up to 30% of Patagonian blenny specimens contained plastics (Pozo et al. 2019), while in southern Patagonia microplastic particles (PET, acrylic, polypropylene, nylon) have been detected in limpets, squat lobsters and southern king crabs (Andrade et al. 2025a). Because these taxa are keystone components in our network, pollutants ingested by them may propagate through trophic links, diminishing prey nutritional quality and transferring contaminants to higher consumers. Such disruptions increase the likelihood of cascading effects and secondary extinctions (Olmo-Gilabert et al. 2024), ultimately undermining ecosystem stability and resilience (Eskuche-Keith et al. 2023).

Another emerging physical stressor is the alteration of salinity due to brine discharge from desalination plants associated with the expected expansion of green hydrogen production in the Strait of Magellan, a region considered strategic for its abundant renewable energy resources (Vicuña et al. 2022). However, the ecological consequences of brine discharge in this unique environment remain largely unstudied, with no official environmental impact assessments available. Our analysis revealed a small-world pattern in the Strait of Magellan's food web, characterized by high clustering and short path lengths among species. While salinity fluctuations are expected to directly affect physiologically sensitive taxa, the mentioned food web pattern suggests that even tolerant species could face indirect impacts due to their trophic proximity to affected nodes, where localized disturbances could quickly propagate, altering species interactions and community structure. Evidence from Patagonian fjords shows that salinity shifts can drive species turnover and reduce benthic diversity (Cari et al. 2020), as well as impair larval fish growth and feeding (Landaeta et al. 2012), underscoring the potential for ecological disruption. Therefore, urgent research on desalination brine impacts in the Strait is needed to ensure that green hydrogen and its supporting infrastructure develop sustainably in this ecologically important region.

Lastly, the intensification of artisanal fishing and the potential opening of new fisheries targeting key species pose significant challenges to ecosystem stability in the Strait of Magellan. According to our results, the Patagonian blenny is a central mid-trophic species, serving both as consumer and as prey for fish, seabirds, and marine mammals. Its overexploitation could trigger cascading effects, altering trophic interactions and reducing food web resilience. In low-productivity systems, the extraction of trophically central species such as forage fish or mid-level predators can erode network cohesion and increase vulnerability to additional stressors (Paine 1966, Jackson et al. 2001, Estes et al. 2011). The loss of redundancy in energy pathways further concentrates ecological functions in a few species, heightening the risk that their decline could lead to disproportionate ecosystem change (Biggs et al. 2020, Sepúlveda et al. 2024). Empirical evidence shows that food webs with limited connectivity and redundancy are more

prone to cascading responses when the abundance of key species fluctuates (Dunne et al. 2002b, Marina et al. 2018a, 2018b, Mérillet et al. 2022). Consequently, establishing new fisheries without considering trophic interactions could exacerbate imbalances, compromising both ecosystem resilience and recovery capacity.

Caveats and future perspectives

The food web analyzed integrates data from the western, central, and eastern micro-basins of the Strait of Magellan, reflecting spatial heterogeneity previously reported in oceanographic studies (Antezana 1999). Despite this variability, a unified web was constructed, supported by the strong longitudinal connectivity linking the Pacific and Atlantic Oceans and reinforcing the Strait's role as a bio-oceanic corridor (Brun et al. 2020). Environmental fluctuations further shape community composition, particularly zooplankton (Guglielmo et al. 2011, Zagami et al. 2011), though benthic responses remain poorly studied (but see Andrade et al. 2016, Hüne et al. 2018). Similar basin-level differences in trophic dynamics have been observed in the Beagle Channel (Giesecke et al. 2021, Bruno et al. 2023, Latorre et al. 2023). These variations likely impact the transfer of energy and matter within the food web. Acknowledging these complexities, we constructed a food web representing year-round dynamics in the Strait of Magellan ecosystem. Further research is needed to better understand how environmental heterogeneity, species traits, and methodological approaches shape the trophic architecture – that is, the structural and functional organization of species interactions and energy flow – in Sub-Antarctic marine food webs, and how this architecture relates to ecosystem stability. Given that this is the first study of its kind in the area, future work should incorporate key environmental factors such as depth, current flow, and nutrient availability, as demonstrated in recent studies on food web structure (Cordone et al. 2020).

A key limitation of this analysis lies in the uneven taxonomic resolution. Several nodes represent broad categories (e.g. Polychaeta, Benthic decapoda, Copepoda), each encompassing ecologically distinct species (Montiel et al. 2011, Aldea et al. 2020). Such aggregation influences both the number of nodes and the density of connectors: while reducing taxonomic resolution lowers the apparent species richness of the network, it simultaneously inflates structural metrics such as connectance and omnivory, because heterogeneous species are collapsed into a single unit with multiple trophic links (Thompson and Townsend 2000, Gauzens et al. 2013). This is particularly critical in the Strait of Magellan, a region that hosts high species richness – 166 polychaetes (Montiel et al. 2011), 412 mollusks (Aldea et al. 2020), and 225 malacostracan crustaceans (Lancellotti and Vasquez 2000) – including endemic taxa such as certain polychaetes (Moreno et al. 2006). Consequently, some nodes may overestimate ecological centrality simply because they artificially concentrate diverse roles.

Nonetheless, this aggregation also reflects methodological constraints: predator diet studies often identify polychaetes

only to class level due to digestion-related limitations. Thus, the Polychaeta node mirrors both the trophic scale of published data and the difficulty of prey identification, though its apparent prominence should be interpreted with caution.

Quantitative data (e.g. abundance, biomass) were not incorporated into the present model, as it is based on binary trophic interactions. However, the availability of such information for certain taxa (Mutschke and Tharje 1999, Retamal and Gorny 2001, Montiel et al. 2011) underscores its importance for future applications of models that explicitly account for interaction strength and energy flux. Addressing this gap will allow a more accurate estimation of species' ecological roles and enhance the predictive capacity of food web analyses.

Additionally, kelp forests – highly abundant in the Strait and known to provide shelter, nursery grounds, and energy sources for a wide range of organisms (Cárdenas et al. 2007, Ríos et al. 2007, Andrade et al. 2016) – were not explicitly emphasized in terms of their topological roles within the food web. This omission stems from the scarcity of updated dietary data, particularly from stomach content analyses, which may lead to underestimating the contribution of macroalgae as a basal energy source in Sub-Antarctic ecosystems. This concern is particularly relevant given the remarkable macroalgal richness reported for the region (Ramírez 2010, Marambio et al. 2016) and the atypical latitudinal gradient where richness increases poleward, contrary to global patterns (Santelices and Marquet 1998). Therefore, future research should prioritize acquiring quantitative dietary data, including prey frequency and occurrence, to improve estimates of interaction strength and better resolve energy pathways in the Strait of Magellan food web.

Building on these considerations, to further advance and benefit understanding of the ecosystem's structure and function, we suggest future research should: 1) assess spatial variability across the Strait's micro-basins, as it may influence food web organization (Kortsch et al. 2019, Cordone et al. 2020); 2) incorporate species traits, such as body size and mass, which play a critical role in predator–prey dynamics (Brose et al. 2019); 3) evaluate the impacts of human-induced stressors as disturbances within the food web (Correa and Gross 2008, da Silva et al. 2023, Giesecke et al. 2024, Salinas et al. 2024); 4) estimate predator–prey interaction strength depicting ecosystem's energy flow by accounting for species traits, habitat structure and population density (Nilsson and McCann 2016, Gauzens et al. 2019); 5) directly test the stability and robustness of the food web under different disturbance scenarios – using, for example, quantitative stability indices such as QSS or simulations of extinctions – based on predator–prey interaction strengths; and 6) assess secondary production, which is key to understanding energy availability for higher trophic levels but remains scarcely quantified in the region, with only few estimates at the population (Andrade et al. 2009) and community levels (Brey and Gerdes 1999). Integrating biomass and production data into network models is essential, since their absence may underestimate the role of species such as the squat lobster

Grimothea gregaria, a dominant prey for many vertebrates and of growing commercial interest for its nutritional value (Varisco et al. 2020), whose biomass reductions could affect apex predators (Haro et al. 2022). Such information would refine energy flow models, improve understanding of benthic–pelagic coupling, and strengthen assessments of ecosystem functional capacity and resilience under future climate and fishing pressures.

Conclusion

The application of an ecological network analysis enabled the identification of species that are crucial for maintaining ecosystem structure in the Strait of Magellan. As the understanding of trophic ecology in Sub-Antarctic marine ecosystems advances, it is essential to integrate these methodological tools into fisheries management policies and marine protected area planning, to ensure a holistic and ecosystem-based approach to conservation – one that explicitly considers the preservation of keystone species and their functional roles within the food web (Christensen et al. 1996). However, current management plans and the designation of protected species often tend to focus on charismatic or commercially important species, potentially overlooking the critical functional roles of less conspicuous organisms within the food web (Boero 2021). Management areas and protected species designations should move beyond a focus solely on charismatic megafauna or target species, and instead more comprehensively consider the underlying ecological structure and functioning of the ecosystem, including the preservation of species that play key roles in maintaining food web stability and resilience (Worm et al. 2006, Long et al. 2015, Shchipanov and Kalinin 2024).

Topological approaches offer valuable insights into the complexity and structure of food webs and their capacity to respond to environmental and anthropogenic stressors (Estrada, 2007, Kortsch et al. 2015, Marina et al. 2024, Scian et al. 2025). By incorporating such an approach, this study contributes to a deeper understanding, at the ecosystem level, of species' role within these 'hidden connections', therefore providing critical information for long-term conservation efforts and resource management in the Strait of Magellan, a sensitive Sub-Antarctic marine region.

As mentioned in Section 'Inference on ecosystem response to stressors', given the growing pressures from climate change and expanding industrial activities in southern Patagonia, including unregulated fisheries, future research should prioritize assessing the resilience of these trophic networks under disturbance scenarios. It should also support the development of adaptive, network-informed management strategies. To effectively address these challenges and to fully realize the potential of the newly defined ecosystemic approach in Chilean law (Ministry of Environment, Resolution 1597 EXENTA 27 March 2025, Law no. 21.600), a paradigm shift is urgently needed. We must move away from traditional species-based management towards approaches that integrate

both the structural and functional properties of ecological networks and the roles of individual species.

Speculations

Our static topological analysis reveals a food web architecture primed for instability. Its low connectance and reliance on a few key hubs suggest that the Strait of Magellan ecosystem may lack the resilience to absorb the pervasive, system-wide pressures of climate change. We speculate that this structural fragility is a harbinger of a potential tipping point. As warming, acidification, and invasive species simultaneously stress multiple nodes, the system's capacity for modular buffering could fail, triggering a rapid, non-linear trophic cascade. This underscores an urgent need to move beyond structural snapshots to quantify the dynamic processes governing the flow of energy and matter. Future ecological research should therefore focus less on compartmentalized analyses and more on frameworks that explicitly embrace interdependence, recognizing ecosystems as integrated biomes whose resilience emerges from connections across scales.

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Data availability statement

Data are available from the Zenodo Digital Repository: <https://doi.org/10.5281/zenodo.17344507> (Andrade et al. 2025b).

Supporting information

The Supporting information associated with this article is available with the online version.

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